

Ibrahim O. Jimoh

Artificial Intelligence & Robotics

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RESEARCH INTERESTS

Human-robot interaction, robot learning and perception, robot manipulation, culturally sensitive social robotics, reinforcement learning, deep learning, and AI safety. Developing intelligent robotic systems that can perceive, learn, and interact appropriately in diverse cultural and social contexts.

EDUCATION

Master of Science, Engineering Artificial Intelligence | July 2023 – May 2025

Carnegie Mellon University (CMU) | Pittsburgh, USA

Specializations:

- Machine Learning, Deep Learning, Robotics, Large Language Models, Reinforcement Learning

Relevant Coursework:

- Applied Stochastic Processes, Machine Learning for Engineers, Introduction to Deep Learning, Cognitive Robotics, Data Inference and Applied Machine Learning, AI System Design, Principles and Applications of AI, AI/ML Operations, Humanoid Robotics and Cognition, Research Methods

Bachelor of Engineering, Computer Engineering | September 2015 – September 2021

Federal University of Technology (FUT) | Minna, Nigeria

Thesis: Development of a Drowsiness Detection System for Road Accident Reduction and Control using Computer Vision Techniques

RESEARCH EXPERIENCE

Research Associate | June 2025 – Present

AI & Robotics Laboratory, Culturally Sensitive Social Robotics | CMU

- Developing a comprehensive cultural knowledge base structured into a cultural ontology to guide robot interactions in culturally attuned manners across diverse African contexts.
- Implementing advanced robot localization software using visual landmarks, triangulation, and trilateration methods to enable precise spatial awareness for social robot navigation and interaction.

Research Assistant | June 2024 – May 2025

AI & Robotics Laboratory, Culturally Sensitive Social Robotics | CMU

- Conducted ethnographic research to identify culturally sensitive robot behaviors and interaction patterns socially acceptable in African contexts, informing the design of culturally-aware robotic systems.
- Developed and implemented a Behavior Tree Domain Specific Language (DSL) for structured human-robot interaction scenarios, enabling flexible and maintainable robot behavior design.
- Created multi-modal robot localization system combining odometry, IMU data, ArUco marker detection, and YOLO-based landmark recognition for robust pose estimation in dynamic environments.

Research Assistant | January 2021 – September 2021

Computer Engineering Laboratory | Federal University of Technology

- Investigated computer vision algorithms for vehicle safety systems, conducting comprehensive surveys on vehicular accident statistics and causal factors.
- Developed in-vehicle drowsiness detection system using Histogram of Oriented Gradient (HOG) facial predictors and eye aspect ratio monitoring with real-time alarm capabilities and emergency notification features.

WORK EXPERIENCE

Software Engineer Intern | January 2023 – June 2023

Optimus Bank | Lagos, Nigeria

- Led as project lead, coordinating timelines, ensuring deliverable completion, and fostering collaboration between development and business units for critical AI integration projects.
- Served in scrum master capacity, facilitating daily stand-ups, sprint planning, and retrospectives, resulting in significant improvements in team productivity and task completion rates.
- Designed and developed RESTful APIs for customer support virtual assistant, enabling seamless integration with existing banking infrastructure and improving service efficiency.
- Maintained rigorous data security protocols and ensured 100% regulatory compliance with banking security standards and industry requirements.

IT Engineer Intern | January 2022 – December 2022

Protogy Global Services Limited | Ibadan, Nigeria

- Managed and maintained electrical meter programming software, implementing updates and optimizing utility performance across production systems.
- Enhanced production efficiency by over 60% through development of Python-based inventory management application, streamlining product entry workflows.
- Resolved critical software and hardware issues in meter pipeline systems, ensuring operational continuity and product quality.
- Recognized as the most efficient and best performing intern for outstanding contributions to operational improvements.

Hardware and Software Technician | May 2016 – May 2019

Canaanland Technologies | Ilorin, Nigeria

- Maintained, troubleshooted, and repaired computer systems and devices for diverse corporate and individual clients, ensuring high customer satisfaction.
- Provided comprehensive technical support and customized software solutions tailored to client needs and specifications.
- Developed strong customer relations skills through direct client interactions and technical consultation services.

PUBLICATIONS

- A. Akinade, D. Barros, E. Birhan, M. Danso, Y. Haile, **I. Jimoh**, C. Osano, P. Ranchod, M. Richard, B. Rosman, B. Shimelis, T. Tefferi, and D. Vernon, "Culturally Sensitive Social Robotics for Africa," chapter in Cultural Robotics Book: Diversified Sustainable Practices, Springer Nature [Under Publication 2025]
- **I. Jimoh**, H. Equbay, C. Osano, T. Tefferi, and D. Vernon, "Behavior Trees for Culturally Sensitive Social Robots: African Culture Case Study," Abstract in Robotics in Africa Forum at IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024.
- A. Akinade, D. Barros, M. Danso, Y. Haile, E. Birhan, B. Shimelis, C. Osano, P. Ranchod, M. Richard, B. Rosman, **I. Jimoh**, T. Tefferi, and D. Vernon, "Culturally Sensitive Social Robotics for Africa," Poster Presentation in Cultural Robotics: Diversified Sustainable Practices, Workshop at IEEE/ACM Human Robot Interaction (HRI) Conference, 2025.
- Y. U. Murph, A. Avoka, **I. Jimoh**, and F. Rurangwa, "Breaking Mental Health Silence: Men from Different Cultures and Generations," Case Studies in International Education, Vol. 5 No. 2, Dec. 2024.

SELECTED RESEARCH PROJECTS

Deep Reinforcement Learning for Mobile Robot Social Navigation | Sep 2024 – Dec 2024

- Developed deep RL framework jointly modeling human-robot and human-human interactions in dynamic social spaces.
- Implemented self-attention architecture capturing long-range dependencies for complex crowd interaction modeling.

LLM-Based Investment Advisor using RAG | Sep 2024 – Dec 2024

- Built scalable LLM-powered platform democratizing financial advising with risk profiling, asset allocation, portfolio forecasting, and personalized recommendations using Retrieval-Augmented Generation.

Deepfake Voice Detection for Fraud Prevention | Jun 2024 – Aug 2024

- Developed and trained deep CNN model to distinguish authentic and synthetic voice recordings with high accuracy.

Robot Arm Manipulation for Structured Tasks | Mar 2024 – Apr 2024

- Developed control strategies for robotic arm pick-and-place tasks demonstrating precise manipulation in structured and semi-structured environments.

Synchronizing Multimodal Behaviors - Eva Robot Head | Sep 2023 – Dec 2023

- Built responsive humanoid robot head using Unity, VR, 3D printing (Fusion 360, Ultimaker Cura).
- Implemented facial expression recognition and mimicking using neural networks; enhanced open-source control code.

TEACHING & MENTORSHIP

Teaching Assistant – Robotics Principles and Practice | Fall 2025

Carnegie Mellon University College of Engineering

- Supported graduate students in robotics concepts, kinematics, motion planning, and control.
- Assisted with lab setup, grading, and detailed feedback on hands-on robotics assignments.

AI Alignment Facilitator | Aug 2025 – Oct 2025

European Network for AI Safety

- Facilitated weekly international discussion groups on AI safety and alignment as part of AI Safety Collab initiative.
- Simplified complex technical concepts and fostered collaboration on AI governance and safety topics.

TECHNICAL SKILLS

Programming Languages:

- Python, C++, C#, Java, JavaScript

AI & Machine Learning:

- PyTorch, TensorFlow, Scikit-Learn, OpenAI API, RAG, LangChain, Pandas, NumPy

Robotics:

- ROS, Gazebo, OpenAI Gym, Robot Kinematics, Motion Planning, Control Systems, Sim2Real

Cloud & DevOps:

- AWS, GCP, Docker, Kubernetes

Developer Tools:

- Git, GitHub, Linux/Ubuntu, Unity3D, LaTeX, Arduino, Raspberry Pi

HONORS & AWARDS

- **HRI Conference Grant Award** | ACM SIGAI, 2025 | Competitive grant to present research at HRI Conference in Melbourne, Australia
- **Mastercard Foundation Scholarship** | 2023 | Highly competitive master's degree scholarship for African students
- **Best School Representative** | International Mathematics Olympiad (IMO), 2015 | Recognized for exemplary performance

CONFERENCE PRESENTATIONS

- **IEEE/ACM HRI Conference** | Melbourne, Australia, 2025 | Poster: "Culturally Sensitive Social Robotics for Africa" at Cultural Robotics Workshop
- **IEEE/RSJ IROS Conference** | Abu Dhabi, UAE, 2024 | Abstract: "Behavior Trees for Culturally Sensitive Social Robots" at Robotics in Africa Forum
- **Touch Sensing and Processing Summer School (TSPSS)** | Dresden, Germany, 2024 | Hands-on training in tactile sensing and manipulation with Meta AI, TouchLab, Stanford, TU Dresden

LEADERSHIP & SERVICE

- **Communications Lead** | Afrisnet Graduate Committee, Jun 2024 – Present | Organizing workshops and career development sessions for 500+ African graduate scholars
- **Interim Chair & Vice Chair** | IEEE Student Branch, CMU-Africa, May 2024 – May 2025 | Coordinated branch activities, organized workshops, led IEEEExtreme 18.0 participation
- **Community Engagement Manager** | Research Club, CMU-Africa, Dec 2023 – Dec 2024 | Established faculty-student research collaboration, organized research workshops and panels
- **Volunteer** | Solid Africa, Dec 2023 | Provided aid and distributed relief materials to 100+ vulnerable hospital patients in Rwanda

PROFESSIONAL MEMBERSHIPS

- Member, Association for Computing Machinery (ACM)
- Graduate Member, Institute of Electrical and Electronics Engineers (IEEE)
- Graduate Member, IEEE Robotics and Automation Society (IEEE-RAS)
- Graduate Member, Nigeria Society of Engineers (NSE)
- Member, Google Developers Group (Active since 2020)